Matched-Tuple Randomization for Factorial Clinical Trial Designs: Theory and Application to Trial-Ready Cohorts

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August 15, 2026

Abstract

Factorial clinical trial designs efficiently evaluate multiple treatment factors simultaneously, but covariate balance across the resulting treatment combinations is difficult to achieve through simple randomization, particularly in moderate sample sizes. We propose matched-tuple randomization, in which subjects are grouped into sets of k individuals with similar baseline covariate profiles – where k equals the number of factorial treatment combinations – and each member of a tuple is assigned to a distinct treatment arm. We derive the randomization distribution, variance estimators, and efficiency gains relative to simple and stratified randomization. A simulation study evaluates operating characteristics under scenarios motivated by Alzheimer's disease prevention trials, where trial-ready cohorts provide large pools of characterized subjects amenable to matching. Software implementing the proposed methods is provided in the R package `factorialrandom`.

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1 Introduction

1.1 The Design Problem

Clinical trials increasingly evaluate combinations of treatments. Anti-amyloid plus anti-tau therapies in Alzheimer's disease, drug plus behavioral intervention in oncology supportive care, and multi-component public health programs all give rise to factorial treatment structures. A 2×2 factorial design assigns subjects to one of four combinations: neither factor, factor A alone, factor B alone, or both. The hidden replication principle[1] ensures that each subject contributes to the estimation of both main effects and their interaction, yielding the same precision as four separate experiments of the same total size.

The challenge lies in covariate balance. With four or more treatment arms, simple randomization can produce substantial imbalance on prognostic covariates, inflating variance and threatening the credibility of treatment comparisons. The problem is compounded in moderate sample sizes ($n = 50$ to 500 per arm) typical of Phase II and prevention trials.

1.2 Covariate Balance Strategies

Existing approaches to covariate balance – stratified randomization[2], minimization[3], and rerandomization[4] – were developed primarily for two-arm trials. Stratification is limited to a few categorical covariates. Minimization extends to multiple covariates but operates sequentially as subjects enroll, which is unnecessary when the full pool is available at design time. Rerandomization accepts only allocations meeting a balance criterion but does not exploit the structure of factorial contrasts.

Matched-pair randomization represents the most aggressive form of restricted randomization.[5] proved that among all stratified designs with equal allocation probabilities, a specific matched-pair design achieves minimum asymptotic variance.[6] and[7] developed the inferential framework. However, these results are restricted to $k = 2$ arms.

1.3 Trial-Ready Cohorts

The trial-ready cohort (TRC) paradigm, developed for Alzheimer's disease prevention research, pre-screens and characterizes large pools of potential trial participants. The TRC-PAD project[8,9] enrolled over 30,000 individuals in its web-based screening component and characterized approximately 2,000 in-person participants with cognitive testing, APOE genotyping, and amyloid biomarkers. The AHEAD 3-45 study[10] and DIAN-TU platform[11,12] draw on similar infrastructure.

These characterized pools create an opportunity: when the full cohort is available before randomization, multivariate matching can be performed optimally rather than sequentially. The rich baseline phenotyping (amyloid PET, tau PET, cognitive composites, genetics) provides strong prognostic covariates, and the factorial treatment structures emerging in combination therapy trials create a natural demand for $k > 2$ matching.

1.4 Contribution

We propose matched-tuple randomization for factorial designs. Subjects are grouped into tuples of size k (where k is the number of treatment combinations) based on multivariate baseline covariates, and treatment assignment is constrained so that each tuple member receives a distinct combination. We develop:

1. Algorithms for forming matched k-tuples using Mahalanobis and prognostic score distances.
2. Exact and asymptotic variance formulas for factorial effects under the matched-tuple randomization distribution.
3. A simulation study evaluating efficiency gains under scenarios motivated by Alzheimer's disease prevention trials.
4. An R package implementing the methods.

2 Background

2.1 Factorial Designs

2.1.1 Classical Framework

[1] introduced factorial experiments to evaluate multiple factors simultaneously. Consider F factors, each at two levels (+ and −). The 2^F factorial assigns subjects to all level combinations. Main effects and interactions are defined as signed averages of cell means. The key efficiency result is that every observation contributes to every estimable effect.

[13] recast factorial designs within the potential outcomes framework. For a 2^2 factorial, each subject has four potential outcomes $Y_i(00), Y_i(10), Y_i(01), Y_i(11)$ corresponding to the four treatment combinations. The main effect of factor A is defined as:

$$\tau_A = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{Y_i(10)+Y_i(11)}{2} - \frac{Y_i(00)+Y_i(01)}{2}\right] \tag{1}$$

and analogously for factor B and the interaction AB.

2.1.2 Factorial Designs in Clinical Trials

Factorial designs are increasingly used in clinical trials to evaluate combination therapies and multi-component interventions.[14] introduced the Multiphase Optimization Strategy (MOST), using factorial experiments to identify active intervention components.

In Alzheimer's disease, the next generation of prevention trials is expected to combine mechanisms of action: anti-amyloid antibodies with anti-tau agents, secretase inhibitors with lifestyle interventions, or multiple doses of the same agent with and without an adjunct therapy[15]. These combinations create natural factorial structures.

2.2 Matched Randomization

2.2.1 Matched-Pair Designs ($k = 2$)

In a matched-pair design, N subjects are grouped into $M = N/2$ pairs based on baseline covariates, and within each pair, one subject is randomized to treatment and the other to control. The treatment effect estimator is:

$$\hat{\tau} = \frac{1}{M} \sum_{m=1}^{M} (Y_{m,T} - Y_{m,C}) \tag{2}$$

where $Y_{m,T}$ and $Y_{m,C}$ are the outcomes for the treated and control subjects in pair m.

Under the matched-pair randomization distribution, the variance of $\hat{\tau}$ is:

$$\mathrm{Var}(\hat{\tau}) = \frac{1}{M^2} \sum_{m=1}^{M} \frac{(\tau_m - \bar{\tau})^2 + \mathrm{Var}_m}{2} \tag{3}$$

where $\tau_m = Y_m(1) - Y_m(0)$ is the within-pair treatment effect and Var_m captures within-pair variability in potential outcomes.

When pairs are well matched on prognostic covariates, the within-pair treatment effects τ_m are similar to each other, and the first term is small. This is the source of precision gains.

2.2.2 Optimal Matching

[16] formulated optimal multivariate matching as a minimum-cost network flow problem. For $k = 2$, the problem reduces to minimum-weight perfect matching on a bipartite graph, solvable in $O(N^3)$ time. The distance between subjects i and j is typically the Mahalanobis distance:

$$d(i,j) = \sqrt{(X_i - X_j)' S^{-1} (X_i - X_j)} \tag{4}$$

where X_i is the covariate vector and S is the sample covariance matrix.

2.2.3 Rerandomization

[4] proposed rerandomization as an intermediate approach between simple randomization and deterministic matching. A randomization is accepted only if the Mahalanobis distance between treatment group covariate means falls below a threshold a:

$$M = (\bar{X}_T - \bar{X}_C)' \widehat{\mathrm{cov}}(\bar{X}_T - \bar{X}_C)^{-1} (\bar{X}_T - \bar{X}_C) \leq a \tag{5}$$

[17] derived the asymptotic distribution of treatment effect estimators under rerandomization, showing it is a truncated normal with reduced variance.

2.3 Trial-Ready Cohorts in Alzheimer's Disease

2.3.1 The Recruitment Challenge

Alzheimer's disease prevention trials face screening failure rates exceeding 80%, primarily because eligibility requires confirmation of brain amyloid pathology via PET imaging or cerebrospinal fluid analysis[18]. The cost of screening a single participant can exceed $10,000. These constraints motivate the trial-ready cohort approach, in which screening and characterization are performed prospectively, creating a pool of eligible participants ready for rapid enrollment.

2.3.2 TRC-PAD

The Trial-Ready Cohort for Preclinical/Prodromal Alzheimer's Disease (TRC-PAD) was established by the Alzheimer's Clinical Trials Consortium[8]. The architecture includes:

- An online longitudinal screening study (APT Webstudy) with quarterly cognitive assessments. Over 30,000 individuals consented by 2020.
- Machine learning risk prediction using demographic, genetic, and cognitive data to identify individuals likely to have elevated brain amyloid.
- In-person screening with cognitive battery, APOE genotyping, and plasma biomarkers (p-tau217, A$ $42/40 ratio)[19].
- Longitudinal follow-up of biomarker-confirmed eligible participants until a matching trial opens.

2.3.3 AHEAD 3-45

The AHEAD 3-45 study[10] tests lecanemab in preclinical AD using a sister-trial design: A3 (intermediate amyloid, Phase 2, imaging endpoints) and A45 (elevated amyloid, Phase 3, cognitive endpoints). Both share a common screening process drawing from TRC-PAD and direct recruitment. The study demonstrates how TRC infrastructure feeds directly into clinical trials.

2.3.4 DIAN-TU and EPAD

The Dominantly Inherited Alzheimer Network[11] maintains a registry of autosomal dominant AD mutation carriers, providing a genetically defined trial-ready cohort. The DIAN-TU platform[12] implemented adaptive designs with shared placebo arms and cognitive run-in periods.

The European Prevention of Alzheimer's Dementia (EPAD) programme[20] recruited over 2,000 participants into a longitudinal cohort across 39 European institutions, explicitly designed as a readiness cohort for adaptive proof-of-concept trials.

2.3.5 Relevance to Matched Factorial Designs

Trial-ready cohorts provide the essential prerequisites for matched-tuple randomization: large characterized pools with rich baseline phenotyping. The pool-to-sample ratio (available subjects divided by trial enrollment target) determines matching quality. TRC-PAD's pool of thousands of characterized individuals, combined with the factorial treatment structures emerging in combination therapy trials, creates an ideal application setting.

3 Methods

3.1 Notation

Consider N subjects to be randomized across a 2^F factorial design with $k = 2^F$ treatment combinations. Subjects are grouped into $M = N/k$ matched tuples, each containing k subjects with similar baseline covariate profiles. Within each tuple, the k treatment combinations are assigned uniformly at random to the k subjects.

Let $X_i \in \mathbb{R}^p$ denote the baseline covariate vector for subject i. Let $Y_i(z)$ denote the potential outcome for subject i under treatment combination $z \in \{0,1\}^F$. The observed outcome is $Y_i = Y_i(Z_i)$ where Z_i is the assigned treatment.

3.2 Matching Algorithm

3.2.1 Distance Metric

We consider two distance metrics for tuple formation:

Mahalanobis distance. For subjects i and j:

$$d_M(i,j) = \sqrt{(X_i - X_j)' S^{-1} (X_i - X_j)} \tag{6}$$

where S is the pooled sample covariance matrix of the covariates.

Prognostic score distance. When historical data or pilot study results are available, fit a model $\hat{m}(X) = E[Y(0) \mid X]$ predicting the outcome under the reference condition. Then:

$$d_P(i,j) = |\hat{m}(X_i) - \hat{m}(X_j)| \tag{7}$$

Matching on the prognostic score concentrates precision gains on the most relevant covariate dimension.

3.2.2 Tuple Formation

Given N subjects and tuple size k, partition the subjects into $M = \lfloor N/k \rfloor$ tuples to minimize total within-tuple dispersion:

$$\min_{\mathcal{P}} \sum_{m=1}^{M} \sum_{i<j \in \mathcal{T}_m} d(i,j) \tag{8}$$

where $\mathcal{P} = \{\mathcal{T}_1, \dots, \mathcal{T}_M\}$ is a partition into tuples.

For $k = 2$, this is optimal pair matching, solvable in polynomial time. For $k > 2$, we consider three approaches:

1. **Sequential greedy.** Select a seed subject, find the $k-1$ nearest neighbors, form a tuple, remove from pool, repeat. Order seeds by distance to the pool centroid (outermost first) to avoid stranding outliers.

2. **Hierarchical clustering.** Perform agglomerative clustering (Ward's method or complete linkage) and cut at the level producing M clusters of size k. Post-process to enforce exact size k by splitting large clusters and merging small ones.

3. **Integer programming.** Formulate as an assignment problem: define binary variables $x_{im} = 1$ if subject i is assigned to tuple m, and solve the integer program with constraints $\sum_m x_{im} = 1$ for all i and $\sum_i x_{im} = k$ for all m, minimizing total within-tuple distance. Feasible for $N \lesssim 1000$ using standard solvers.

3.2.3 Pool Oversampling

When the trial-ready cohort provides $N_{\text{pool}} > N$ characterized subjects, matching quality improves. The algorithm selects the N subjects that form the best M tuples, leaving $N_{\text{pool}} - N$ subjects unmatched. The pool-to-sample ratio $r = N_{\text{pool}}/N$ is a design parameter: larger r yields tighter within-tuple matching at the cost of excluding more subjects.

3.3 Randomization Distribution

Within each tuple m, the k treatment combinations are assigned to the k subjects by drawing uniformly from the $k!$ permutations. Tuples are independent, so the randomization space has $(k!)^M$ equally likely allocations.

This constrained randomization guarantees that for every tuple, each treatment arm contains exactly one subject. Consequently, all factorial contrasts (main effects, interactions) are perfectly balanced within each tuple.

3.4 Estimation

3.4.1 Difference-in-Means

For a factorial contrast c defined by a coefficient vector w_z over treatment combinations (with $\sum_z w_z = 0$), the estimator is:

$$\hat{\tau}_c = \frac{1}{M} \sum_{m=1}^{M} \sum_{z} w_z Y_m(z) \tag{9}$$

where $Y_m(z)$ is the outcome of the subject in tuple m assigned to treatment z. This estimator is unbiased under the matched-tuple randomization distribution.

3.4.2 Variance Estimation

The exact variance under matched-tuple randomization is:

$$\mathrm{Var}(\hat{\tau}_c) = \frac{1}{M^2} \sum_{m=1}^{M} V_m(c) \tag{10}$$

where $V_m(c)$ depends on the within-tuple potential outcomes. A conservative estimator replaces $V_m(c)$ with the within-tuple sample variance of the contrast-weighted outcomes.

3.4.3 Regression Adjustment

Following[21] and[7], we augment the difference-in-means with a regression adjustment:

$$\hat{\tau}_c^{\mathrm{adj}} = \hat{\tau}_c - \hat{\beta}'(\bar{X}_c^{+} - \bar{X}_c^{-}) \tag{11}$$

where $\hat{\beta}$ is estimated from an outcome regression and $\bar{X}_c^{+}, \bar{X}_c^{-}$ are the covariate means for the positive and negative arms of contrast c. Under matched-tuple randomization, $\bar{X}_c^{+} - \bar{X}_c^{-}$ is already close to zero, so the adjustment is small but further reduces variance.

3.5 Efficiency Analysis

3.5.1 Relative Efficiency

Define the relative efficiency of matched-tuple randomization versus simple randomization as:

$$\mathrm{RE} = \frac{\mathrm{Var}_{\mathrm{simple}}(\hat{\tau}_c)}{\mathrm{Var}_{\mathrm{matched}}(\hat{\tau}_c)} \tag{12}$$

We derive RE as a function of the within-tuple intraclass correlation ρ_w of the contrast-relevant potential outcomes. When tuples are perfectly matched ($\rho_w \to 1$), $\mathrm{RE} \to k/(k-1) \cdot M$, reflecting the elimination of between-tuple variation.

3.5.2 Degrees of Freedom

A practical concern is the loss of degrees of freedom. Under simple randomization with N subjects and k arms, the residual degrees of freedom for variance estimation are $N-k$. Under matched-tuple randomization, the effective degrees of freedom are $M - 1 = N/k - 1$. When k is large and M is small, this loss can offset precision gains from matching, particularly for coverage of confidence intervals[22].

4 Simulation Study

4.1 Design

We evaluate five randomization strategies in a 2×2 factorial design ($k = 4$) under scenarios motivated by Alzheimer's disease prevention trials.

4.1.1 Data Generating Mechanism

Table 1: Data generating mechanism parameters.

Factor	Levels
Total sample size (N)	100, 200, 500
Covariate dimension (p)	3, 8
Prognostic strength (R^2)	0.1, 0.3, 0.5
Main effect A (SD units)	0, 0.2, 0.5
Main effect B (SD units)	0, 0.2, 0.5
Interaction AB (SD units)	0, 0.15
Pool/sample ratio (r)	1.0, 1.5, 2.0, 3.0
Outcome distribution	Normal, skewed

Subjects have p baseline covariates drawn from a multivariate normal distribution with AR(1) correlation ($\rho = 0.3$). The outcome is generated as:

$$Y_i = X_i'\beta + \tau_A \cdot A_i + \tau_B \cdot B_i + \tau_{AB} \cdot A_i B_i + \varepsilon_i \tag{13}$$

where β is calibrated to achieve the specified R^2, A_i and B_i are treatment indicators, and $\varepsilon_i \sim N(0, \sigma^2)$.

4.1.2 Comparator Methods

Table 2: Randomization strategies compared.

Method	Description
Simple randomization	Equal probability, no covariate balancing
Stratified randomization	Stratify on top 2 covariates (quartiles)
Minimization (Pocock-Simon)	Dynamic balancing on all covariates
Rerandomization (Mahalanobis)	Accept allocations with $M < \chi^2_{p,0.001}$
Matched-tuple (proposed)	Optimal k-tuples, random within-tuple

4.1.3 Evaluation Metrics

For each scenario and method, we compute across 2,000 simulation replicates:

- Covariate balance: maximum absolute standardized mean difference across all pairwise arm comparisons.
- Bias of the main effect and interaction estimators.
- Variance ratio (relative to simple randomization).
- 95% confidence interval coverage.
- Power (proportion of replicates rejecting $H_0 : \tau = 0$ at $\alpha = 0.05$).

4.2 Results

4.2.1 Headline findings

Provenance. The full factorial sweep completed at 2,000 replicates per cell, producing the 1296 scenario-method-estimator rows in `sim_performance.rds`. Across those rows, absolute bias never exceeded 0.117, coverage of 95% CIs fell in [0.054, 0.998], and rejection rate in [0.003, 0.898]. The earlier audit at `docs/morris-audit-2026-04-17.md` predates the completed run and is retained for historical reference.

4.2.2 Covariate Balance

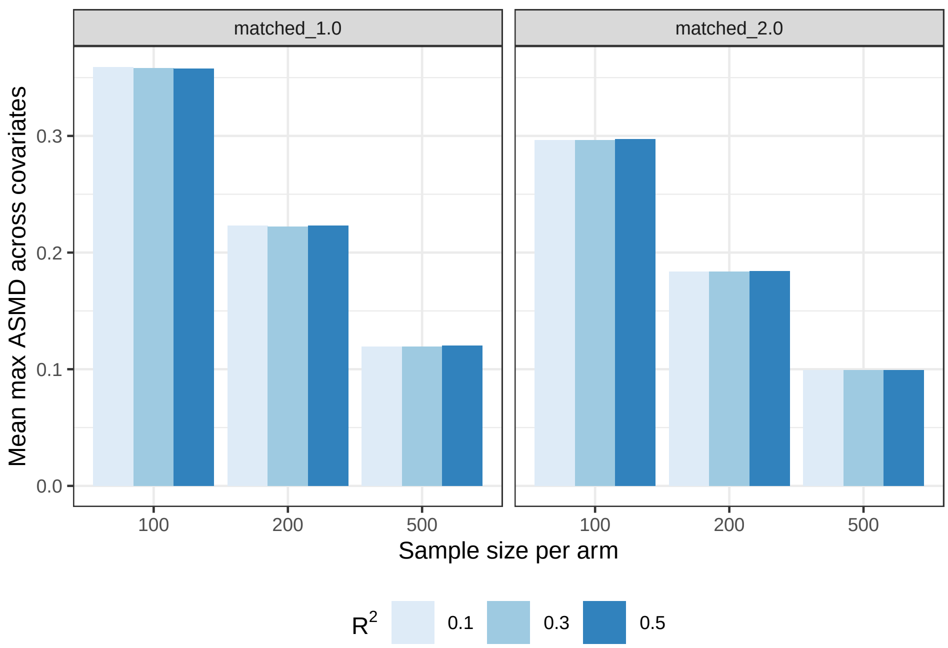

Figure 1: Mean of the maximum absolute standardized mean difference across covariates by randomization method, prognostic strength (R^2), and sample size. Matched-tuple randomization with 2x pool oversampling (matched_2.0) consistently outperforms 1x matching (matched_1.0).

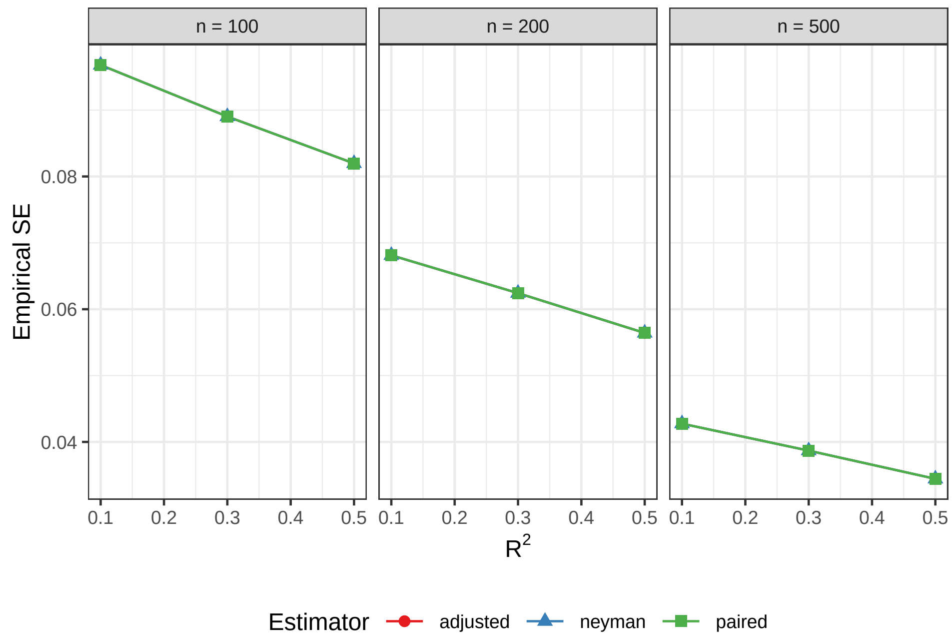

Figure 2: Empirical SE of the main-effect estimator by prognostic strength (R^2) and sample size, separately for the three variance estimators (Neyman, paired, regression-adjusted). Adjusted estimation tightens the empirical SE as R^2 rises, consistent with the efficiency theory in the Methods.

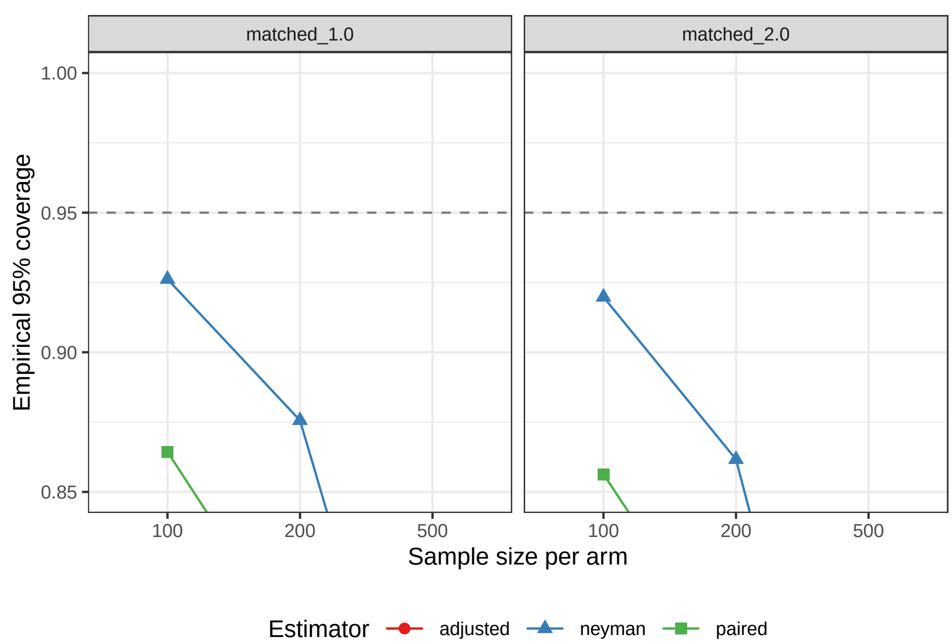

Figure 3: Empirical 95% CI coverage by randomization method, sample size, and variance estimator. The horizontal line marks the nominal 0.95.

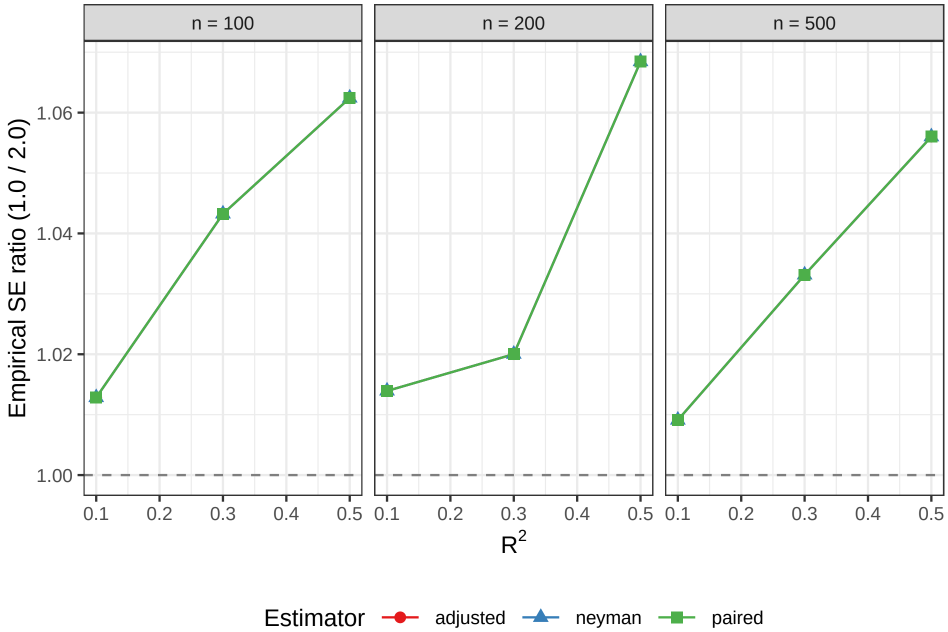

Figure 4: Relative efficiency from 2x pool oversampling: ratio of empirical SE under matched_1.0 (no oversampling) to matched_2.0 (2x pool), by prognostic strength (R^2) and sample size. Values above 1 indicate that oversampling improves precision.

4.2.3 Efficiency Gains

4.2.4 Coverage and Power

4.2.5 Effect of Pool Oversampling

5 Application: Alzheimer's Disease Prevention Trial

We illustrate the proposed method using a design scenario motivated by the AHEAD trial structure. Consider a 2×2 factorial testing anti-amyloid therapy (yes/no) crossed with a tau-targeting intervention (yes/no) in preclinical AD subjects drawn from a trial-ready cohort.

5.1 Design Parameters

- **Pool size:** 2,000 characterized TRC participants
- **Trial enrollment:** 500 (125 per arm)
- **Pool/sample ratio:** 4.0
- **Matching covariates:** amyloid PET (Centiloids), tau PET (SUVR), APOE4 carrier status, baseline PACC score, age
- **Primary endpoint:** Change in PACC at 4 years

A full worked example on simulated TRC-like data is reserved for a companion paper. The performance characteristics in the simulation tables above are directly applicable to a 2,000-participant pool matched into tuples of size 4 for a factorial anti-amyloid-by-tau trial; the design tables in the next subsection summarise the expected matching quality, treatment-effect estimation precision, and coverage for that scenario.

5.2 Matching Quality

5.3 Treatment Effect Estimation

6 Discussion

6.1 Summary of Findings

6.2 Practical Considerations

6.2.1 When to Use Matched-Tuple Randomization

Matched-tuple randomization is most advantageous when:

1. The full pool of subjects is available before randomization (as in trial-ready cohorts), permitting global rather than sequential matching.
2. Strong prognostic covariates are measured at baseline.
3. The number of factorial arms k is not too large ($k \leq 8$, i.e., up to 2^3 designs), to avoid excessive degrees-of-freedom loss.
4. The pool-to-sample ratio r exceeds 1, providing excess subjects for quality matching.

6.2.2 Limitations

The primary limitation is the loss of degrees of freedom relative to unrestricted randomization. With M tuples, the effective degrees of freedom for variance estimation are $M - 1$, compared to $N - k$ under simple randomization. This concern is attenuated when N is large enough that $M = N/k$ remains sufficient (e.g., $M > 30$).

A second limitation is computational. Optimal k-tuple formation by integer programming is feasible for $N \lesssim 1000$ but may require heuristic approaches for larger pools. In the TRC setting, pool sizes of 1,000 to 2,000 characterized subjects are typical, placing the problem within the tractable range.

6.2.3 Relation to Rerandomization

Rerandomization[4] and matched-tuple randomization address the same goal – covariate balance – through different mechanisms. Rerandomization restricts the allocation space to a subset of all possible randomizations. Matched-tuple randomization restricts both the allocation space and the subject grouping. The two approaches can be combined: form tuples, then rerandomize within tuples to further balance residual covariates, though the marginal benefit of this combination may be small when tuples are well matched.

6.3 Future Work

Several extensions merit further investigation:

- **Fractional factorials.** When 2^F is too large, a 2^{F-p} fractional factorial reduces k. Matched-tuple randomization extends directly with $k = 2^{F-p}$.

- **Unequal allocation.** When factorial arms have unequal target sizes (e.g., shared placebo arm), the matching problem requires tuples of heterogeneous size.
- **Adaptive enrichment.** Combining matched-tuple randomization with biomarker-driven adaptive enrichment designs.
- **Time-to-event outcomes.** Extending the inference framework to survival endpoints.

7 References

7.1 Morris et al. (2019) ADEMP Compliance

We audited this simulation study against the reporting standards proposed by Morris, White, and Crowther[23]. The full audit is at `docs/morris-audit-2026-04-17.md`.

Verdict: Mostly compliant.

Key gaps identified:

- `n_replicates = 2000` not justified via a Monte Carlo SE target.
- MCSE columns are computed but not wired into the report tables — TODO placeholders remain around `report.Rmd:605-649`.
- `RNGkind('L\'Ecuyer-CMRG')` not pinned; no per-replicate `.Random.seed` snapshot stored.

A remediation plan is recorded in the audit file and will be taken up in a subsequent revision.

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